Differential Equations

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Differential Equations

DEFINITIONS :

An equation which involves independent and dependent variables and their derivatives is called a **DIFFERENTIAL EQUATION.**

e.g. $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 0$

Every differential equation represents a family of curves. To find that family is to solve that differential equation and its solution is also known as its PRIMITIVE.

Order and Degree of Differential Equation

Order of a differential equation is known as the order of the highest differential coefficient occurring in it.

Note:

Order of differential equation is positive integer.

Example : $\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 0$ has order 2.

Degree of differential equation is the degree of highest order derivative occuring in it

To find degree differential equation should be expressed as polynomial in derivatives and it should be free from radicals and fractions.

$$f(x, y)\left[\frac{d^m y}{dx^m}\right]^p + \phi(x, y)\left[\frac{d^{m-1}(y)}{dx^{m-1}}\right]^q + \ldots = 0$$

is order m and degree p.

e.g.,(i) $\frac{d^2y}{dx^2} = \left[y + \left(\frac{dy}{dx}\right)^6\right]^{1/4}$ ⟨ order 2, degree 4

(ii) $\frac{dy}{dx} + y = \frac{1}{\frac{dy}{dx}}$ ⟨ order 1, degree 2

(iii) $e^{\frac{d^3y}{dx^3}} - x\frac{d^2y}{dx^2} + y = 0$ ⟨ order 3, (not applicable)

(iv) $\sin^{-1}\left(\frac{dy}{dx}\right) = x + y \Rightarrow \frac{dy}{dx} = \sin(x + y)$ ⟨ order 1, degree 1

(v) $\ln\left(\frac{dy}{dx}\right) = ax + by \Rightarrow \frac{dy}{dx} = e^{ax + by}$ ⟨ order 1, degree 1

Q. **The order and degree of the differential equation $\frac{d^2y}{dx^2} = \left\{1 + \left(\frac{dy}{dx}\right)^4\right\}^{5/3}$ are,**

respectively,

(A) 2, not applicable **(B) 1, 4** **(C) 2, 3** **(D) 2, 5**

Sol. **(C)**

$$\frac{d^2y}{dx^2} = \left\{1 + \left(\frac{dy}{dx}\right)^4\right\}^{5/3}$$

$$\Rightarrow \left(\frac{d^2y}{dx^2}\right)^3 = \left\{1 + \left(\frac{dy}{dx}\right)^4\right\}^5$$

Hence, the order is 2 and the degree is 3.

Q. **Find order and degree of the differential equation $\frac{d^3y}{dx^3} = x \ln\left(\frac{dy}{dx}\right)$ are,**

respectively,

(A) 3, not applicable **(B) 3, 1** **(C) 1, 3** **(D) not applicable, 2**

Sol. **(A)**

Clearly, the order is 3 and the degree is not defined due to $\ln\left(\frac{dy}{dx}\right)$ term.

FORMING OF A DIFFERENTIAL EQUATION

To remove arbitrary constants from a differential equation, we follow following steps

(I) We differentiate given equation w.r.t independent variable. It should be noted that equation should be differentiated as many times as number of arbitrary constants in it.

(II) Use above equations obtained to remove arbitrary constants

(III) Equations we get in terms of derivative free from arbitrary constants is required differential equations

Examples:

Q. **Tell differential equation of all concentric circles having center at the origin**

Sol. Equation of such circles $x^2 + y^2 = r^2$
Differentiating $xdx + ydy = 0$
Note that the order is 1.

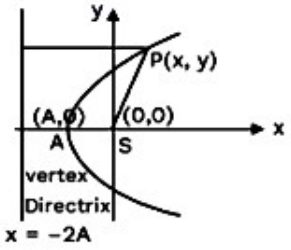


Q. **What will be the differential equation of the family of parabolas with focus at origin and axis of symmetry along the x-axis.**

Sol. Equation of the parabola is
$(x^2 + y^2) = (x + 2A)^2$
$y^2 = 4A\,(A + x)$

$$2y\frac{dy}{dx} = 4A$$

$$\therefore y\frac{dy}{dx} = 2A$$



Hence $y^2 = \left(y\frac{dy}{dx}\right)^2 + 2y\frac{dy}{dx}.x$ or $y^2 = y^2\left(\frac{dy}{dx}\right)^2 + 2xy\frac{dy}{dx}$ — order 1, degree 2

Q. **Form the differential equation whose general solution is $y = Ae^x + Be^{-x}$ (A, B are arbitrary constants).**

Sol. We have
$y = Ae^x + Be^{-x}$ ...(i)
Differentiating w.r.t.x we get

$$\frac{dy}{dx} = Ae^x - Be^{-x}$$

$$\text{and } \frac{d^2y}{dx^2} = Ae^x + Be^{-x} = y \quad \text{...(ii)}$$

From Equations (i) and (ii) we have

$$\frac{d^2y}{dx^2} - y = 0$$

Which is the differential equation whose general solution is $y = Ae^x + Be^{-x}$,

.

Q. **The differential equation of all ellipses having axes along co-ordinate axes, is:**

(A) $x\left(\frac{dy}{dx}\right)^2 = y\frac{dy}{dx} + xy\frac{d^2y}{dx^2}$ **(B)** $y\left(\frac{dy}{dx}\right)^2 + xy\frac{d^2y}{dx^2} = x\frac{dy}{dx}$

(C) $x\left(\frac{dy}{dx}\right)^2 + xy\frac{d^2y}{dx^2} = y\frac{dy}{dx}$ **(D) None of these**

Sol. **(C)**

Equation of such ellipse is

$ax^2 + by^2 = 1$

Differentiating w.r.t.x, we get

$2ax + 2by\,y_1 = 0$

$\Rightarrow ax + byy_1 = 0 \Rightarrow \frac{-a}{b} = \frac{yy_1}{x}$ ...(i)

Again differentiating w.r.t. x, we get

$\Rightarrow a + by_1^2 + byy_2 = 0 \Rightarrow \frac{-a}{b} = y_1^2 + yy_2$ ...(ii)

From equations (i) and (ii), we get

$\frac{yy_1}{x} = y_1^2 + yy_2$

$\Rightarrow yy_1 = xy_1^2 + xyy_2$

Q. **What will be the differential equation of all parabolas whose axis are parallel to the y-axis is**

(A) $\frac{d^3y}{dx^3} = 0$ **(B)** $\frac{d^2x}{dy^2} = C$ **(C)** $\frac{d^3y}{dx^3} + \frac{d^2x}{dy^2} = 0$ **(D)** $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = 0$

Sol. **(A)**

The equation of a member of the family of parabolas having axis parallel to y-axis is

$y = Ax^2 + Bx + C$ ...(1)

where A, B, and C are arbitary constants

Differentiating equatin (1) w.r.t. x, we get $\frac{dy}{dx} = 2Ax + B$ ...(2)

Which on again differentiating w.r.t. x gives $\frac{d^2y}{dx^2} = 2A$ ...(3)

Differentiating (3) w.r.t.x, we get $\frac{d^3y}{dx^3} = 0$

Q. **The differential equation of the family of circles with fixed radius 5 units and centre on the line y = 2 is**

(A) $(x-2)^2 y'^2 = 25 - (y-2)^2$

(B) $(x-2) y'^2 = 25 - (y-2)^2$

(C) $(y-2) y'^2 = 25 - (y-2)^2$

(D) $(y-2)^2 y'^2 = 25 - (y-2)^2$

Sol. **(D)**

The equation of circle is

$(x-\alpha)^2 + (y-2)^2 = 25$ ...(1)

Differentiating w.r.t. x, we get

$$(x-\alpha) + (y-2)\frac{dy}{dx} = 0 \quad \text{...(2)}$$

From (1) and (2) on eliminating 'α', we get

$$(y-2)^2\left(\frac{dy}{dx}\right)^2 + (y-2)^2 = 25$$

$\Rightarrow (y-2)^2(y')^2 = 25 - (y-2)^2$

GENERAL AND PARTICULAR SOLUTIONS

A solution of differential equations which contains arbitrary constants of same number equal to order of differential equation is known as **general solution**

and when we assign some value to those arbitrary constants we get a particular solution

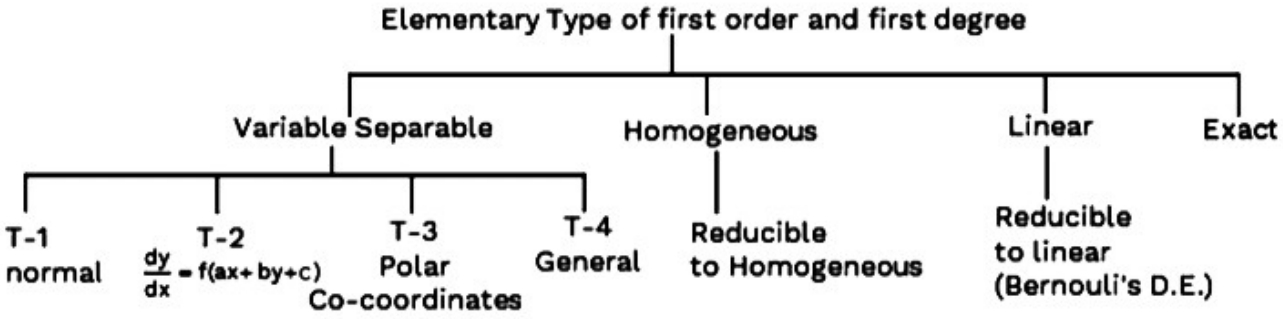
order of differential equations = number of independent arbitrary constants

By independent it means we should not be able to make differential equation of similar relation containing fewer arbitrary constants

For example $y = m\, e^{x+n}$ are not independent as the equation can be written as $y = m\, e^n . e^x = ke^x$. Similarly $y = (C_1 + C_2)\cos(x + C_3) - C_4 e^{x+C_5}$

has only 3 independent arbitrary constants hence will be of order 3.

TYPES OF FIRST ORDER & FIRST DEGREE DIFFERENTIAL EQUATION:



Variables Separable :

Differential equation which can be expressed as ;

f(x) dx + g(y) dy = 0 is said to be variable - separable type

VARIABLE SEPARABLE (Type-1) :

A general solution will be given by $\int f(x)dx + \int g(y)dy = c$;

where c is the arbitrary constant.

Examples :

Q. $\ln \frac{dy}{dx} = ax + by$

Sol. $\ln \frac{dy}{dx} = ax + by \Rightarrow \frac{dy}{dx} = e^{ax+by} = e^{ax} \cdot e^{by}$

or $\int e^{-by}\, dy = \int e^{ax}\, dx$

or $\frac{e^{-by}}{-b} = \frac{e^{ax}}{a} + c$

Q. **Solve the equation $\cos^2 y \cot x\, dx + \cos^2 x \cot y\, dy = 0$**

Sol. The given equation can be rewritten as

$$\int \frac{\sec^2 x}{\tan x} dx + \int \frac{\sec^2 y}{\tan y} dy = \log c$$

$\Rightarrow$ log tan x + log tan y = log c; where c is an arbitrary positive constant

$\Rightarrow$ tan x tany = c .

Q. **The equation of the curve passing through the point (1, 1) and satisfying the differential equation $\frac{dy}{dx} = e^{x-y} + x^2e^{-y}$ is given by**

(A) $e^y = e^x + \frac{x^2}{2} - \frac{1}{2}$ **(B)** $e^y = e^x + \frac{x}{3} - \frac{1}{3}$

(C) $e^y = e^x + \frac{x^3}{3} - \frac{1}{3}$ **(D)** $e^y = e^x - \frac{x}{3} + \frac{1}{3}$

Sol. **(C)**

We have

$\frac{dy}{dx} = e^{x-y} + x^2e^{-y}$

$\Rightarrow e^y dy = (e^x + x^2)\, dx$ (Variables separable)

Integrating we get

$\int e^y dy = \int \left(e^x + x^2\right) dx$

$\Rightarrow e^y = e^x + \frac{1}{3}x^3 + c$

The curve passes through (1, 1) implies

$e = e + \frac{1}{3} + c \Rightarrow c = -\frac{1}{3}$

Therefore the curve equation is

$e^y = e^x + \frac{x^3}{3} - \frac{1}{3}$

Q. **Solve $\log \frac{dy}{dx} = 4x - 2y - 2$, given that y = 1 when x= 1.**

Sol. Given $\log \frac{dy}{dx} = 4x - 2y - 2$

$\Rightarrow \frac{dy}{dx} = e^{4x-2y-2}$

$\Rightarrow \int e^{2y+2} dy = \int e^{4x} dx$

$\Rightarrow \frac{e^{2y+2}}{2} = \frac{e^{4x}}{4} + c$

$x = 1, y = 1 \Rightarrow \frac{e^4}{2} = \frac{e^4}{4} + c$ or $c = e^4 / 4$

$$\Rightarrow \quad \frac{e^{2y+2}}{2} = \frac{e^{4x}}{4} + \frac{e^4}{4}$$

$$\Rightarrow \quad 2e^{2y+2} = e^{4x} + e^4$$

Q. **The family of curves (not passing through origin) for which the length of the normal at any point equals the length of the radius vector joining the point with the origin is (are).**

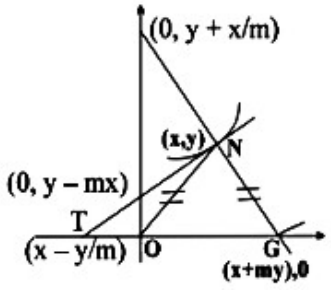
(A) circles or ellipses centre at origin

(B) ellipses and rectangular hyperbolas with centre at origin

(C) rectangular hyperbolas and circles with centre at origin

(D) Pair of lines and ellipses (centre origin)

Sol. **(C)**



Let y =f(x) be the curve. Length of the normal at N(x, y) is ON.

Given ON = GN

$$\left| y\left(1+\left(\frac{dy}{dx}\right)^2\right)^{1/2} \right| = \sqrt{x^2+y^2}$$

$$\Rightarrow y^2\left(1+\left(\frac{dy}{dx}\right)^2\right) = x^2 + y^2$$

$$\Rightarrow y^2\left(\frac{dy}{dx}\right)^2 = x^2 \Rightarrow y\frac{dy}{dx} = \pm x$$

Case I: $y\frac{dy}{dx} = x$. Then

$y\,dy = x\,dx$ (Variables separable)

$$\Rightarrow \frac{1}{2}y^2 = \frac{1}{2}x^2 + c$$

$\Rightarrow y^2 - x^2 = 2c$

Which is a rectangular hyperbola.

Case II : $y\frac{dy}{dx} = -x$. Then

$ydy + xdx = 0$

$\Rightarrow y^2 + x^2 = 2c$

Which is a circle.

Q. **If for some curve the tangent to it at any point P intersects the coordinate axis at A and B. If P is the mid-point of AB then find the curve given that curve passes through (2,3)**

(A) $y^2 + x^2 = 13$ **(B)** $xy = 6$

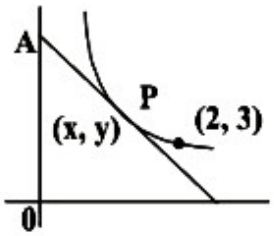
(C) $3x^2 = 4y$ **(D)** $2x^2 + y^2 = 17$

Sol. **(B)**

The equation of tangent at any point P(x,y) is

$$Y - y = \frac{dy}{dx}(X - x) \qquad \ldots(1)$$

$$\text{Gives } x - \text{intercept} = x - y\frac{dx}{dy}$$



As P is mid-point of intercept

$$\frac{x - y\frac{dx}{dy} + 0}{2} = x$$

$$\Rightarrow \quad x - y\frac{dx}{dy} = 2x$$

$$\Rightarrow -\frac{dy}{y} = \frac{dx}{x}$$

Integrating we get xy=c

Since the curve passes through (2, 3), c = 6.

$\Rightarrow$ xy = 6

Q. **If the curve passing through (1, 2) for which the segment of the tangent between P and T is bisected at its point of intersection with the y-axis is a conic, then its length of latus rectum.**

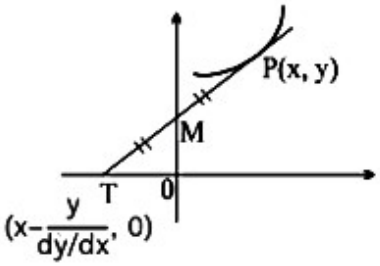
(A) 64 **(B) 16** **(C) 4** **(D) 8**

Sol. **(C)**

The equation of the tangent at any point P(x,y) is

$$Y - y = \frac{dy}{dx}(X - x) \qquad \ldots(1)$$

$$\text{Gives } x - \text{intercept} = x - y\frac{dx}{dy}$$



As Mis mid-point of P and T

$$\frac{x - y\frac{dx}{dy} + x}{2} = 0$$

$$\Rightarrow \quad y\frac{dx}{dy} = 2x$$

$$\Rightarrow \frac{2dy}{y} = \frac{dx}{x}$$

Integrating we get $2 \ln y = \ln x + \ln c$

$\Rightarrow y^2 = cx$

Since the curve passes through (1, 2), $c = 4$.

$\Rightarrow y^2 = 4x$

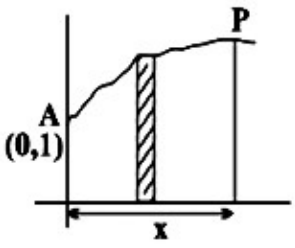
MULTI-CORRECT TYPE

Q. **Let the curve $y = f(x)$ is such that area bounded by the curve, the co-ordinate axes and a variable ordinate is equal to the length of the corresponding arc of curve. Given that the curve passes through (0, 1) then f(2) can be**

(A) 1 **(B) 2** **(C)** $\frac{e^2 + e^{-2}}{2}$ **(D)** $\frac{e^2 - e^{-2}}{2}$

Sol. **(AC)**

$$\therefore \int_0^x y\,dx = \int_A^P dl = \int \sqrt{(dx)^2 + (dy)^2} = \int_0^x \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\,dx$$



Now differentiate both sides.

$$y = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$$

$$\Rightarrow \quad \frac{dy}{dx} = \pm\sqrt{y^2 - 1}$$

Case I $\quad y \neq \pm 1$

$$\frac{dy}{\sqrt{y^2 - 1}} = \pm dx$$

Integrating

$$\ln\left(y + \sqrt{y^2 - 1}\right) = \pm x + c$$

through (0, 1) $\Rightarrow c = 0$

$$\ln\left(y + \sqrt{y^2 - 1}\right) = \pm x$$

$$\left(y + \sqrt{y^2 - 1}\right) = e^{\pm x}$$

$$\left(\sqrt{y^2 - 1}\right) = e^{\pm x} - y$$

$-1 = e^{\pm 2x} - 2ye^{\pm x}$

$$y = \frac{e^x + e^{-x}}{2}$$

Case II : $y = \pm 1$

Here $y = 1$ is satisfying differential equation.

Q. **A certain radiactive substance is present M_1 grams at time t_1 and M_2 grams at time t_2 then find it's half time.**

Sol. Let k be decay constant

$$\frac{dy}{dt} = -ky$$

$$\Rightarrow \quad \frac{dy}{y} = -kdt$$

$$\Rightarrow \quad \ln y = -kt + c$$

$$\Rightarrow \quad y = e^{-kt+c}$$

$$\Rightarrow \quad y = e^c e^{-kt}$$

Let $e^c = M_0$

$$\Rightarrow \quad y = M_0 e^{-kt} \qquad ...(1)$$

Now, $M_1 = M_0 e^{-kt_1}$ ...(2)

and $M_2 = M_0 e^{-kt_2}$ ...(3)

$\frac{(2)}{(3)}$ gives

$$\frac{M_1}{M_2} = e^{-k(t_1 - t_2)}$$

$$\Rightarrow k = \frac{\ln(M_1 / M_2)}{t_2 - t_1}$$

Half pefiod = $t_{1/2}$

$$t_{1/2} = \frac{\ln 2}{k}$$

$$t_{1/2} = (t_2 - t_1)\frac{\ln 2}{\ln(M_1 / M_2)}$$

VARIABLE SEPARABLE (Type-2) :

$\frac{dy}{dx} = f(ax + by + c)$, $b \neq 0$. (If $b = 0$ this is directly variable separable)

For solving this , substitute $t = ax + by + c$. After which equation reduces to separable type in the variable t and x which can be solved.

EXAMPLES

Q. **Solve** $(x + y)^2 \frac{dy}{dx} = a^2$

Sol. Let $x + y = t$

$\Rightarrow 1 + \frac{dy}{dx} = \frac{dt}{dx}$

so given equation becomes

$t^2\left(\frac{dt}{dx} - 1\right) = a^2$

$\Rightarrow \quad \frac{t^2}{a^2 + t^2} dt = dx$

$\Rightarrow \quad \int\left(1 - \frac{a^2}{t^2 + a^2}\right)dt = \int dx$

$\Rightarrow \quad t - a\tan^{-1}\left(\frac{t}{a}\right) = x + c$

$\Rightarrow \quad x + y - a\tan^{-1}\left(\frac{x + y}{a}\right) = x + c$

$\Rightarrow \quad y = a\tan^{-1}\left(\frac{x + y}{a}\right) + c$

Q. **Solve** $\frac{dy}{dx}\sqrt{1 + x + y} = x + y - 1.$

Sol. Putting $\sqrt{1 + x + y} = v$, we have

$\Rightarrow \quad x + y - 1 = v^2 - 2$

$\Rightarrow \quad 1 + \frac{dy}{dx} = 2v\frac{dv}{dx}$

Then the given equation transforms to

$\left(2v\frac{dv}{dx} - 1\right)v = v^2 - 2$

$$\Rightarrow \quad \frac{dv}{dx} = \frac{v^2 + v - 2}{2v^2}$$

$$\Rightarrow \quad \int \frac{2v^2}{v^2 + v - 2} dv = \int dx$$

$$\Rightarrow \quad 2\int \left[1 + \frac{1}{3(v-1)} - \frac{4}{3(v+2)}\right] dv = \int dx$$

$$\Rightarrow \quad 2\left[v + \frac{1}{3}\log|v-1| - \frac{4}{3}\log|v+2|\right] = x + c$$

where $v = \sqrt{1 + x + y}$

Q. **Solve** $\frac{dy}{dx} = \frac{2x + 3y - 1}{4x + 6y - 5}$

Sol. Let $2x + 3y - 1 = v$

$$\Rightarrow \quad 2 + 3\frac{dy}{dx} = \frac{dv}{dx}$$

So, gives equation becomes

$$\frac{\frac{dv}{dx} - 2}{3} = \frac{v}{2v - 3}$$

$$\Rightarrow \quad \frac{dv}{dx} = \frac{3v}{2v - 3} + 2$$

$$\Rightarrow \quad \frac{2v - 3}{7v - 6} dv = dx$$

$$\Rightarrow \quad \frac{14v - 21}{14v - 12} dv = \frac{7}{2} dx$$

$$\Rightarrow \quad \frac{(14v - 12) - 9}{14v - 12} dv = \frac{7}{2} dx$$

$$\Rightarrow \quad \left(1 - \frac{9}{14v - 12}\right) dv = \frac{7}{2} dx$$

$$\Rightarrow \quad v - \frac{9}{14}\ln|14v - 12| = \frac{7}{2}x + c$$

$\Rightarrow$ $14v - 9\ln|14v - 12| = 49x + K$,

Where $v = 2x + 3y - 1$

Q. **If curves passing through (0, 0) satisfies $\frac{dy}{dx} = (x + y)^2$ then it is represented by:**

(A) $\tan^{-1}(x + y) = x$ **(B) $\tan^{-1}(x + y) = y$**
(C) $\tan^{-1}(x) = x + y$ **(D) $\tan^{-1}(y) = x + y$**

Sol. **(A)**

$$\frac{dy}{dx} = (x + y)^2 \qquad ...(i)$$

Here the variable are not separable but by putting $x + y = v$, we have

$$1 + \frac{dy}{dx} = \frac{dv}{dx}$$

$\Rightarrow$ Equation (i) reduces to

$$\frac{dv}{dx} = v^2 + 1 \text{ or } \int \frac{dv}{v^2 + 1} = \int dx \qquad ...(ii)$$

$\tan^{-1} v = x + c$
As it passes through origin
$\tan^{-1}(x + y) = x$ which is a required solution.

VARIABLE SEPARABLE (Type – 3)

$$\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2} \qquad \text{where } b_1 + a_2 = 0$$

Approach:
(Simply cross multiply and note the perfect differential of xy, Integrate term by term)

Q. **If solution curves of $\frac{dy}{dx} = \frac{4x - 3y}{3x - 2y}$ don't pass through origin then it can be**

(A) Ellipse **(B) Parabola** **(C) Hyperbola** **(D) Pair of lines**

Sol. **(C)**

Cross multiply
$3xdy - 2ydy = 4xdx - 3ydx$
or $3(xdy + ydx) - 2y\,dy - 4xdx = 0$
or $3\,d(xy) - 2ydy - 4xdx = 0$;
Integration gives $3xy - y^2 - 2x^2 = c$
As $h^2 > ab$. It is hyperbola.

Q. **Solve :** $\frac{dy}{dx} = \frac{x-2y+5}{2x+3y-1}$

Sol. Cross multiplying we get

$2xdy + 3ydy - dy = xdx - 2ydx + 5dx$

$2(xdy + ydx) + 3ydy - dy - xdx - 5dx = 0$

$2(d(xy)) + 3ydy - dy - xdx - 5dx = 0$

integrating we get

$2xy + 1.5y^2 - y - 0.5x^2 - 5x = c$

$4xy + 3y^2 - 2y - x^2 - 10x = 2c$

POLAR COORDINATES (Type – 4)

Many a times transformating to the polar co-ordinates easies separation of variables. For which we should remember the following differentials.

(A) If $x = r\cos\theta$; $y = r\sin\theta$ where r and θ both are variable.

(i) $x\,dx + y\,dy = r\,dr$ **(ii)** $x\,dy - y\,dx = r^2\,d\theta$

(B) If $x = r\sec\theta$ & $y = r\tan\theta$

(i) $x\,dx - y\,dy = r\,dr$ **(ii)** $x\,dy - y\,dx = r^2\sec\theta\,d\theta$.

Proof of A :

$x = r\cos\theta$; $y = r\sin\theta$

$\Rightarrow x^2 + y^2 = r^2$ and $\tan\theta = y/x$

Hence , $x\,dx + y\,dy = rdr$ and $xdy - y\,dx = x^2\sec^2\theta\,d\theta$

$xdy - ydx = r^2\,d\theta$

Proof of B:

$x = r\sec\theta$ and $y = r\tan\theta$

$\Rightarrow x^2 - y^2 = r^2$ and $y/x = \sin\theta$

Hence $xdx - ydy = rdr$ and $xdy - ydx = x^2\cos\theta\,d\theta$

$xdy - ydx = r^2\sec\theta\,d\theta$

Q. **Solve : $(x + xy)dx = (x^2 - y)dy$**

Sol. $x = r\cos\theta$; $y = r\sin\theta$; $\tan\theta = \frac{y}{x}$

$\Rightarrow$ $x\,dx + y\,dy = r\,dr$; $xdy - ydy = r^2\,d\theta$

$\therefore$ $rdr = r\cos\theta\, r^2\,d\theta$

$$\int \frac{dr}{r^2} = \int \cos\theta\,d\theta$$

$\Rightarrow \quad -\frac{1}{r} = \sin\theta + C \qquad$ or $\qquad \frac{1}{\sqrt{x^2+y^2}} + \frac{y}{\sqrt{x^2+y^2}} = C$

$y + 1 = C\sqrt{x^2+y^2} \qquad$ or $\qquad (y+1)^2 = C^2(x^2+y^2)$

Q. **Solve :** $\frac{xdx+ydy}{\sqrt{x^2+y^2}} = \frac{ydx-xdy}{x^2}$

Sol. Put $x = r\cos\theta$; $y = r\sin\theta$

$\Rightarrow \quad x^2 + y^2 = r^2 \Rightarrow xdx + ydy = rdr$

& $\quad \frac{y}{x} = \tan\theta \qquad \Rightarrow \qquad \frac{xdy-ydx}{x^2} = \sec^2\theta\, d\theta$

$\therefore \quad \int\frac{rdr}{r} = -\int\sec^2\theta\, d\theta$

$r = -\tan\theta + C$

$\sqrt{x^2+y^2} + \frac{y}{x} = C$

Q. **The solution of** $\frac{xdx+ydy}{xdy-ydx} = \sqrt{\frac{1-x^2-y^2}{x^2+y^2}}$ **is**

(A) $\sqrt{x^2+y^2} = \sin\left\{\tan^{-1}(y/x) + C\right\}$ **(B)** $\sqrt{x^2+y^2} = \cos\left\{\left(\tan^{-1}y/x\right) + C\right\}$

(C) $\sqrt{x^2+y^2} = \left(\tan\left(\sin^{-1}(y/x)\right) + C\right)$ **(D) None of these**

Sol. **(A)**

$$\frac{x + y\frac{dy}{dx}}{x\frac{dy}{dx} - y} = \sqrt{\frac{1-x^2-y^2}{x^2+y^2}} \Rightarrow \frac{xdx+ydy}{xdy-ydx} = \sqrt{\frac{1-\left(x^2+y^2\right)}{\left(x^2+y^2\right)}}$$

Put $x = r\cos\theta$, $y = r\sin\theta$ so $xdx + ydy = r\,dr$; $x\,dy - y\,dx = r^2\,d\theta$

$$\Rightarrow \frac{rdr}{r^2d\theta} = \frac{\sqrt{1-r^2}}{r} \Rightarrow \int\frac{dr}{\sqrt{\left(1-r^2\right)}} = \int d\theta \Rightarrow \sin^{-1}(r) = \theta + c$$

$$\Rightarrow r = \sin(\theta + c); \sqrt{\left(x^2+y^2\right)} = \sin(\theta + c) \text{ where } \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

Allternative

The D.E. can be re-written as

$$\frac{xdx+ydy}{\sqrt{1-\left(x^2+y^2\right)}} = \frac{xdy-ydx}{\sqrt{x^2+y^2}}$$

Since $d\tan^{-1}(y/x) = \frac{xdy - ydx}{x^2 + y^2}$,

and $d(x^2 + y^2) = 2(xdx + ydy)$,

$\therefore$ we have $\frac{\frac{1}{2}d(x^2 + y^2)}{\sqrt{x^2 + y^2}\sqrt{1 - (x^2 + y^2)}} = \frac{xdy - ydx}{x^2 + y^2} = d\{\tan^{-1}(y/x)\}$

Put $x^2 + y^2 = t^2$ in the L.H.S and get

$$\frac{tdt}{t\sqrt{1 - t^2}} = d\left\{\tan^{-1}(y/x)\right\}$$

Integrating both sides, we get
$\sin^{-1} t = \tan^{-1}(y/x) + c$

i.e., $\sin^{-1}\sqrt{(x^2 + y^2)} = \tan^{-1}(y/x) + c$

Q. **Solution of** $\frac{xdx - ydy}{xdy - ydx} = \sqrt{\frac{1 + x^2 - y^2}{x^2 - y^2}}$ **is:**

(A) $x^2 - y^2 + \sqrt{(x^2 - y^2) + (x^2 - y^2)^2} = c(x + y)$

(B) $x^2 + y^2 + \sqrt{(x^2 - y^2) + (x^2 - y^2)^2} = c(x - y)$

(C) $x^2 - y^2 + \sqrt{(x^2 + y^2) + (x^2 - y^2)^2} = c(x - y)$

(D) $x^2 - y^2 + \sqrt{(x^2 - y^2) + (x^2 + y^2)^2} = c(x + y)$

Sol. **(A)**

Let $x = r\sec\theta$, $y = r\tan\theta$
So, given equation becomes

$$\frac{rdr}{r^2 \sec\theta d\theta} = \sqrt{\frac{1 + r^2}{r^2}}$$

$$\Rightarrow \frac{dr}{\sqrt{1 + r^2}} = \sec\theta d\theta$$

Integrating

$$\ln\left|r + \sqrt{1 + r^2}\right| = \ln|\sec\theta + \tan\theta| + \ln|c|$$

$$\Rightarrow \quad r + \sqrt{1 + r^2} = c(\sec\theta + \tan\theta)$$

$$\Rightarrow \quad r^2 + r\sqrt{1+r^2} = c(r\sec\theta + r\tan\theta)$$

$$\Rightarrow \quad x^2 - y^2 + \sqrt{\left(x^2-y^2\right) + \left(x^2-y^2\right)^2} = c(x+y)$$

Homogeneous Equations :

A differential equation of the form $\frac{dy}{dx} = \frac{f(x,y)}{\phi(x,y)}$, where f (x , y) & ϕ (x , y) are homogeneous functions of x & y containing the same degree , is called HOMOGENEOUS DIFFERENTIAL EQUATION. This equation can be reduced to the form $\frac{dy}{dx} = g\left(\frac{x}{y}\right)$ & can be solved by putting y = vx changing the dependent variable y to another variable v , where v is some unknown function, which changes differential equation to an equation with variables separable.

Consider $\frac{dy}{dx} + \frac{y(x+y)}{x^2} = 0$.

Important Note :

(a) For function f (x , y) to be a homogeneous function of degree n , it is necessary that for any real number t ($\neq$ 0) , we should have f (tx , ty) = t^n f(x, y).

e.g. $f(x, y) = ax^{2/3} + hx^{1/3} \cdot y^{1/3} + by^{2/3}$ is a homogeneous function of degree 2/3.

(b) A differential equation of the form $\frac{dy}{dx} = f(x, y)$ is homogeneous if f(x , y) is a homogeneous function of degree zero which means f (tx , ty) = t^0 f (x , y) = f(x , y).

Examples on Homogeneous:

Q. **Solve** $\frac{dy}{x+y} + \frac{ydx}{x^2} = 0$

Sol. The given differential equation can be re-written as

$$\frac{dy}{dx} = -\frac{y(x+y)}{x^2} \text{ or } \frac{dy}{dx} = -\frac{y}{x} - \frac{y^2}{x^2}$$

Putting y = vx

$$\Rightarrow \quad \frac{dy}{dx} = v + x\frac{dv}{dx}$$

given equation transforms to

$$v + x\frac{dv}{dx} = -v - v^2$$

$$\Rightarrow \int \frac{dv}{v^2 + 2v} = -\int \frac{dx}{x}$$

$$\Rightarrow \frac{1}{2}\int\left[\frac{1}{v} - \frac{1}{v+2}\right]dv = -\int \frac{dx}{x}$$

$$\Rightarrow \log v - \log(v+2) = -2\log(x) + \log(c)$$

$$\Rightarrow \frac{vx^2}{v+2} = c$$

$$\Rightarrow \frac{x^2 y}{2x + y} = c$$

Q. **Solve** $\left(x\frac{dy}{dx} - y\right)\tan^{-1}\frac{y}{x} = x$ **given** $y|_{x=1} = 0$

Sol. $x\left(\frac{dy}{dx} - \frac{y}{x}\right).\tan^{-1}\frac{y}{x} = x$ or $\left(\frac{dy}{dx} - \frac{y}{x}\right)\tan^{-1}\frac{y}{x} = 1$

put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$ or $\left(v + x\frac{dv}{dx} - v\right)\tan^{-1} v = 1$

$x\frac{dv}{dx}\tan^{-1} v = 1$ or $\int \tan^{-1} v \; dv = \int \frac{dx}{x}$

$v\tan^{-1}v - \int \frac{v}{1+v^2}\,dv = \ln x + C$

$v.\tan^{-1}v - \frac{1}{2}\ln(1+v^2) = \ln x + C$

where $x = 1$; $y = 0 \Rightarrow v = 0 \Rightarrow C = 0$

$$\frac{y}{x}\tan^{-1}\frac{y}{x} = \ln\sqrt{1 + \frac{y^2}{x^2}} + \ln x = \ln x\sqrt{1 + \frac{y^2}{x^2}} = \ln\sqrt{x^2 + y^2}$$

$$\therefore \sqrt{x^2 + y^2} = e^{\frac{y}{x}\tan^{-1}\frac{y}{x}}$$

Q. **Solve differential equation** $y\left(\frac{dy}{dx}\right)^2 + 2x\frac{dy}{dx} - y = 0$; $y|_{x=0} = \sqrt{5}$

Sol. $$\frac{dy}{dx} = \frac{-2x \pm \sqrt{4x^2 + 4y^2}}{2y}$$

$$\frac{dy}{dx} = \frac{-x \pm \sqrt{x^2 + y^2}}{y}$$ which is homogeneous

put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$$v + x\frac{dv}{dx} = \frac{-x \pm x\sqrt{1+v^2}}{vx} = \frac{-1 \pm \sqrt{1+v^2}}{v}$$

$$\therefore\ x\frac{dv}{dx} = \frac{-1 \pm \sqrt{1+v^2}}{v} - v = \frac{-1 \pm \sqrt{1+v^2} - v^2}{v}$$

Taking +ve sign

$$\int \frac{v\,dv}{\sqrt{v^2+1} - (1+v^2)} = \int \frac{dx}{x} \qquad \text{or} \qquad \int \frac{v\,dv}{\sqrt{v^2+1}\left(\sqrt{1+v^2}-1\right)} = -\int \frac{dx}{x}$$

Let $\sqrt{1+v^2} - 1 = t$ $\qquad\qquad$ $\frac{v}{\sqrt{1+v^2}}dv = dt$

$$\therefore\ \int \frac{dt}{t} = -\int \frac{dx}{x}$$

$\ln tx = C$

$$\ln\left[\left(\sqrt{1+\frac{y^2}{x^2}} - 1\right)x\right] = C \quad \text{or} \quad \ln\left[\sqrt{x^2+y^2} - x\right] = C$$

put $x = 0\ ;\ y = \sqrt{5}$

$\ln\left[\sqrt{5} - 0\right] = C \Rightarrow C = \ln\sqrt{5}$

$\therefore\ \sqrt{x^2+y^2} = \sqrt{5} + x$

$x^2 + y^2 = 5 + x^2 + 2\sqrt{5}\,x$

$\therefore\quad y^2 = 5 + 2\sqrt{5}\,x$

Similarly, with −ve sign , $y^2 = 5 - 2\sqrt{5}\,x]$ $y = \sqrt{5 \pm 2\sqrt{5}\,x}$

Q. **If the algebraic sub-tangent at any point of a curve is equal to arithmetic mean of the coordinates of the point, then the equation of the curve is**
(A) $x^2 + y^2 = cy$ (B) $(x-y)^2 = cy$ (C) $(x-y)^2 = cx$ (D) $x^2 - y^2 = c^2$

Sol. **(B)**

By hypothesis

$$\frac{y}{dy/dx} = \frac{x+y}{2}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{2y}{x+y} \text{ (Homogeneous)}$$

Put $y = vx$. Then $v + x\frac{dv}{dx} = \frac{2v}{1+v}$

$\Rightarrow \quad x\frac{dv}{dx} = \frac{2v}{1+v} - v = \frac{v - v^2}{1+v}$

$\Rightarrow \quad \frac{1+v}{v(1-v)}dv = \frac{dx}{x}$

integrating we get

$$\int \frac{1+v}{v(1-v)}dv = \int \frac{dx}{x}$$

$\Rightarrow \quad \int\left(\frac{1}{v} + \frac{2}{1-v}\right)dv = \int \frac{dx}{x}$

$\Rightarrow \quad \log(v) - 2\log(1-v) + \ln c = \log(x)$

$\Rightarrow \quad (1-v)^2 x = cv$

$\Rightarrow \quad (x-y)^2 = cy$

Q. **Consider the curve such that the angle, formed with the x-axis by the tangent to the curve at any of its points, is twice the angle formed by the polar radius of the point of tangency with the x-axis. Such curves are:**

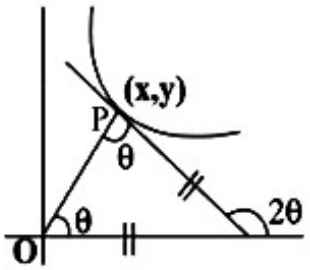
(A) circles touching a fixed line

(B) circles passing through a fixed point

(C) circles having fixed centre

(D) circles passing through a fixed point and which also touches a fixed line

Sol. **(D)**



Given $\tan 2\theta = \frac{dy}{dx}$ where $\tan\theta = \frac{y}{x}$

$$\frac{2\tan\theta}{1-\tan^2\theta} = \frac{dy}{dx}$$

or $\frac{dy}{dx} = \frac{2\frac{y}{x}}{1-\frac{y^2}{x^2}} = \frac{2xy}{x^2-y^2}$ (which is homogeneous)

put $y = vx \Rightarrow \frac{dy}{dx} = v + x\frac{dv}{dx}$

$\therefore \quad v + x\frac{dy}{dx} = \frac{2v}{1-v^2}$

$$\int \frac{1-v^2}{v(1+v^2)}dv = \int \frac{dx}{x}$$

$$\int \frac{1+v^2-2v^2}{v(1+v^2)}dv = \int \frac{dx}{x}$$

$$\int \frac{1+v^2-2v^2}{v(1+v^2)}dv = \int \frac{dx}{x}$$

$$\int \left(\frac{1}{v} - \frac{2v}{(1+v^2)}\right)dv = \int \frac{dx}{x}$$

$$\ln v - \ln(1+v^2) = \ln x + \ln c$$

$$\ln \frac{v}{(1+v^2)} = \ln xc$$

$$\frac{v}{(1+v^2)} = xc$$

$x^2 + y^2 - 2ky = 0$ Ans.

The curve is a circle with center on y-axis and touching the x-axis at the origin.

Equations Reducible To The Homogeneous Form :

If $\frac{dy}{dx} = \frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2}$; where $a_1b_2 - a_2b_1 \neq 0$ and $a_2 + b_1 \neq 0$,

i.e. $\frac{a_1}{b_1} \neq \frac{a_2}{b_2}$

then the substitution of $x = u + h$, $y = v + k$ will transform this equation to a homogeneous type which includes new variables u and v where h and k are constants to be chosen so as to make the given equation homogeneous.

(i) $a_1b_2 - a_2b_1 = 0$, then a substitution of $u = a_1x + b_1y$ will transform the differential equation to an equation with variables separable.

e.g. $\frac{dy}{dx} = \frac{2x + 3y - 1}{4x + 6y - 5}$

(ii) $b_1 + a_2 = 0$, then a simple cross multiplication and substitution of d (xy) in place of x dy + y dx will help in integrating term by term and yields the result easily.

e.g. $\frac{dy}{dx} = \frac{x - 2y + 5}{2x + y - 1}$

(iii) e.g. $\frac{dy}{dx} = \frac{2x - y + 1}{6x - 5y + 4}$

Q. **Solve** $\frac{dy}{dx} = \frac{x - 2y + 5}{2x + y - 1}$.

Sol. The cross multiplying, we get

$2xdy + ydy - dy = xdx - 2ydx + 5dx$

$\Rightarrow \quad 2(xdy + ydx) + ydy - dy = xdx + 5dx$

$\Rightarrow \quad 2d(xy) + ydy - dy = xdx + 5dx$

On integrating, we get $2(xy) + \frac{y^2}{2} - y = \frac{x^2}{2} + 5x + c$

Q. **Solve** $\frac{dy}{dx} = \frac{2x + 3y - 1}{4x + 6y - 5}$

Sol. Let $2x + 3y - 1 = v$

$\Rightarrow \quad 2 + 3\frac{dy}{dx} = \frac{dv}{dx}$

So, gives equation becomes

$$\frac{\frac{dv}{dx} - 2}{3} = \frac{v}{2v - 3}$$

$$\Rightarrow \quad \frac{dv}{dx} = \frac{3v}{2v - 3} + 2$$

$$\Rightarrow \quad \frac{2v - 3}{7v - 6}dv = dx$$

$$\Rightarrow \quad \frac{14v - 21}{14v - 12}dv = \frac{7}{2}dx$$

$$\Rightarrow \quad \frac{(14v - 12) - 9}{14v - 12}dv = \frac{7}{2}dx$$

$$\Rightarrow \quad \left(1 - \frac{9}{14v - 12}\right)dv = \frac{7}{2}dx$$

$$v - \frac{9}{14}\ln|14v - 12| = \frac{7}{2}x + c$$

$14v - 9\ln|14v - 12| = 49x + k$, Where $v = 2x + 3y - 1$

Q. **If a curve satisfies differential equation** $\frac{dy}{dx} = \frac{2x - y + 1}{6x - 5y + 4}$ **has its equation**

$(4y - ax - b) = k(5y - cx - d)^4$ **then the value of a + b + c + d**

(where k is arbitrary constant)

(A) 13 **(B) 14** **(C) 12** **(D) 15**

Sol. **(C)**

$x = u + h$; $y = v + k$; $dx = du$; $dy = dv$

$$\frac{dy}{dx} = \frac{dv}{du}$$

$$\therefore \quad \frac{dv}{du} = \frac{2(u+h)-(v+k)+1}{6(u+h)-5(v+k)+4} = \frac{2u-v}{6u-5v}$$

where $2h - k + 1 = 0$ and $6h - 5k + 4 = 0$ gives $h = -1/4$, $k = 1/2$

put $v = u\,z \Rightarrow \frac{dv}{du} = u\frac{dz}{du} + z$

$$u\frac{dz}{du} + z = \frac{2-z}{6-5z}$$

$$\frac{u\,dz}{du} = \frac{2-z}{6-5z} - z = \frac{2-z-6z+5z^2}{6-5z}$$

$$\Rightarrow \frac{6-5z}{5z^2-7z+2}dz = \frac{du}{u}$$

$$\frac{6-5z}{(5z-2)(z-1)}dz = \frac{du}{u}$$

$$\frac{-20}{3(5z-2)} + \frac{1}{3(z-1)} = \frac{du}{u}$$

Integrating

$$-\frac{4}{3}\ln(5z-2) + \frac{1}{3}\ln(z-1) = \ln u + \frac{\ln c}{3}$$

$$\ln(z-1) = 3\ln u + \ln c + 4\ln(5z-2)$$

$$\ln(v-u) = \ln(c(5v-2u)^4)$$

$$(v-u) = c(5v-2u)^4$$

Put the value of u and v

$$\left(y - x - \frac{3}{4}\right) = c(5y - 2x - 3)^4$$

$$(4y - 4x - 3) = k(5y - 2x - 3)^4 \qquad \text{where } k = 4c$$

Q. **If slope of tangent to curve at any point (x,y) is $\frac{2(y+2)^2}{(x+y-1)^2}$ has its expression in form of $\ln(y+l) + k\tan^{-1}\left(\frac{y+m}{f(x)}\right) = c$, where f(x) is monic polynomial then tell the value of k + l + m + f(4)**

(A) 3 **(B) 4** **(C) 2** **(D) None of these**

Sol. **(D)**

$$\frac{dy}{dx} = \frac{2(y+2)^2}{(x-3+y+2)^2}$$

Let $y + 2 = Y$, $x - 3 = X$

$$\frac{dY}{dX} = \frac{2Y^2}{(X+Y)^2}$$

Let $Y = vX \Rightarrow \frac{dY}{dX} = v + X\frac{dv}{dX}$

$$v + X\frac{dv}{dX} = \frac{2v^2}{(v+1)^2} \Rightarrow X\frac{dv}{dX} = \frac{-v^3 - v}{(v+1)^2}$$

$$\frac{v^2+1+2v}{v(v^2+1)} = -\frac{dx}{x} \Rightarrow \frac{dv}{v} + \frac{2}{v^2+1}dv = -\frac{dX}{X}$$

Integrating $\ln v + 2\tan^{-1} v = -\ln X + c$
(Putting valve of v)

$$\ln\left(\frac{Y}{X}X\right) + 2\tan^{-1}\frac{Y}{X} = c$$

$$\Rightarrow \quad \ln(Y) + 2\tan^{-1}\left(\frac{Y}{X}\right) = c$$

$$\Rightarrow \quad \ln(y+2) + 2\tan^{-1}\left(\frac{y+2}{x-3}\right) = c$$

Linear Differential Equations :

A differential equation is said to be linear if the dependent variable & all its differential coefficients occur in one degree only and are never multiplied together.

The nth order linear differential equation is of the form;

$$a_0(x)\frac{d^n y}{dx^n} + a_1(x)\frac{d^{n-1} y}{dx^{n-1}} + \ldots + a_n(x) \cdot y = \phi(x).$$

Where $a_0(x)$, $a_1(x)$ ..... $a_n(x)$ are called the coefficients of the differential equation.

Linear Differential Equations Of First Order :

The most general form of a linear differential equations of first order is $\frac{dy}{dx} + Py = Q$, where

P & Q are functions of x (Independent variable).

Point to Remember!!!

Note that a linear differential equation is always of the first degree but every differential equation of the first degree need not be linear. e.g. the differential equation $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 + y^2 = 0$ is not linear, though its degree is 1.

In physics it will look like $\frac{ds}{dt} + f(t).s = g(t)$

To solve such an equation multiply both sides by $e^{\int P dx}$ or $e^{\int f(t)dt}$.

Note :

(1) The factor $e^{\int P dx}$ on multiplying by which the left hand side of the differential equation becomes the differential coefficient of some function of x & y, is called integrating factor of the differential equation popularly abbreviated as I. F.

(2) It is very important to remember that on multiplying by the integrating factor, the left hand side becomes the derivative of the product of y and the I.F.

(3) Some times a given differential equation becomes linear if we take y as the independent variable and x as the dependent variable. e.g. the equation;

$(x + 2y^3) = \frac{dx}{dy} y$ can be written as $\frac{dx}{dy} = \frac{x + 2y^3}{y} = \frac{1}{y} x + 2y^2$

which is a linear differential equation.

Q. $\frac{dx}{x + xt^2 - t^2} = \frac{dt}{t(1+t^2)}$; $x|_{t=1} = -\frac{\pi}{4}$

Sol. $\frac{dx}{dt} = \frac{x(1+t^2) - t^2}{t(1+t^2)} = \frac{x}{t} - \frac{t}{1+t^2}$

$\frac{dx}{dt} - \frac{1}{t} x = -\frac{t}{1+t^2}$ (linear differential equation)

I.F. $= e^{-\int \frac{1}{t} dt} = e^{-\ln t} = \frac{1}{t}$

$\therefore \quad \frac{x}{t} = -\int \frac{dt}{1+t^2} = -\tan^{-1} t + c$

$\Rightarrow \quad x = t\,(c - \tan^{-1} t)$

$t = 1 \; ; \; x = -\frac{\pi}{4} \Rightarrow c = 0$

hence $x = -t \tan^{-1} t$ Ans

Q. $x(x^2+1)\frac{dy}{dx} = y(1-x^2) + x^2 \ln x$

Sol. $\frac{dy}{dx} = \frac{y(1-x^2)}{x(1+x^2)} + \frac{x\ln x}{1+x^2}$

$\frac{dy}{dx} - \frac{(1-x^2)y}{x(1+x^2)} = \frac{x\ln x}{1+x^2}$

$\text{I.F.} = e^{\int \frac{x^2-1}{x(x^2+1)}dx} = e^{\int \frac{1-\frac{1}{x^2}}{x+\frac{1}{x}}} = e^{\ln\left(x+\frac{1}{x}\right)} = x + \frac{1}{x}$

$\therefore \quad y.\left(\frac{x^2+1}{x}\right) = \int \frac{x\ln x}{x^2+1}.\frac{x^2+1}{x}dx$

$y\left(\frac{x^2+1}{x}\right) = x(\ln(x) - 1) + c$

Q. **$x\ln x\frac{dy}{dx} + y = 2\ln x$**

Sol. $x\ln x\frac{dy}{dx} + y = 2\ln x$

$\frac{dy}{dx} + \frac{1}{x\ln x}y = \frac{2}{x}$

$\text{I.F.} = e^{\int \frac{1}{x\ln x}dx} = e^{\ln(\ln x)} = \ln x$

$\therefore \quad y\ln x = 2\int \frac{\ln x}{x}dx + c$

$y\ln x = \ln^2 x + c$ Ans.

Q. **If solution of $(x^2-1)\sin x\frac{dy}{dx} + [2x\sin x + (x^2-1)\cos x]y - (x^2-1)\cos x = 0$ is $y(x^2-1)\sin x = f(x)\sin x + g(x)\cos x + c$ then the least value of $f(x) + g(x)$**

(A) -3 **(B) -4** **(C) -2** **(D) -5**

Sol. **(B)**

$\frac{dy}{dx} + \frac{2x\sin x + (x^2-1)\cos x}{(x^2-1)\sin x}y = \cot x$

$\text{I.F.} = e^{\int P(x)dx} = e^{\int \frac{2x\sin x + (x^2-1)\cos x}{(x^2-1)\sin x}dx}$

Let $(x^2 - 1)\sin x = t$

$IF = e^{\int \frac{dt}{t}} = e^{\ln t} = t = (x^2 - 1)\sin x$

Solution is $y(x^2 - 1)\sin x = \int (x^2 - 1)\cos x\,dx + c$

Applying by parts (taking $x^2 - 1$ as I and cosx as II)

$y(x^2 - 1)\sin x = (x^2 - 1)\sin x - \int 2x \sin x\,dx + c$

$y(x^2 - 1)\sin x = (x^2 - 1)\sin x + 2x\cos x - 2\int \cos x\,dx + c$

$y(x^2 - 1)\sin x = (x^2 - 1)\sin x + 2x\cos x - 2\sin x + c$

$y(x^2 - 1)\sin x = (x^2 - 3)\sin x + 2x\cos x + c$

$f(x) + g(x) = x^2 + 2x - 3$

$= (x + 1)^2 - 4$

Q. **Let the function y = f(x) is such that the initial ordinate of any tangent is less than the abscissa of the point of tangency by two units. If f(1) = 0 then which of following doesn't holds good**

(A) f(x) is odd function

(B) function is unbounded

(C) $f(e^2) = -2$

(D) f(x) is aperiodic

Sol. **(A)**

Equation of tangent (x,y) to curve y = f(x) is

$Y - y = f'(x)(X - x)$

Putting X = 0, the initial ordinate of the tangent is

$Y = y - x f'(x)$

$y - x f'(x) = x - 2$

$\Rightarrow \frac{dy}{dx} + \left(-\frac{1}{x}\right)y = \frac{2}{x} - 1$

$I.F. = e^{\int -\frac{dx}{x}} = e^{-\ln x} = \frac{1}{x}$

$y \cdot \frac{1}{x} = \int \left(\frac{2}{x^2} - \frac{1}{x}\right)dx$

$\frac{y}{x} = -\frac{2}{x} - \ln|x| + c$

$y = cx - x\ln|x| - 2$

$y(1) = 0$ gives $c = 2$
$y = 2x - x \ln |x| - 2$
Which is unbounded and aperiodic.

Q. **Solve** $\frac{dy}{dx} = \frac{1}{x\cos y + \sin 2y}$

Sol. $\frac{dy}{dx} = \frac{1}{x\cos y + \sin 2y}$

$\frac{dx}{dy} = x\cos y + \sin 2y$

$\frac{dx}{dy} + (-\cos y)\, x = \sin 2y$

I.F. $= e^{\int -\cos y dy} = e^{-\sin y}$

$\therefore \quad x e^{-\sin y} = \int e^{-\sin y} 2\sin y \cos y \, dy$

Put $\sin y = t$, $\cos y = \frac{dt}{dy}$

$\int e^{-\sin y} 2\sin y \cos y dy = \int e^{-t}(2t)dt = 2\left[-te^{-t} - e^{-t}\right] = -2e^{-t}(t+1) = -2e^{-\sin y}(\sin y + 1)$

$\therefore xe^{-\sin y} = -2e^{-\sin y}(\sin y + 1) + c$
$x = -2(\sin y + 1) + c\, e^{\sin y}$.

Q. **Solution of differential equation** $\frac{dy}{dx} = \frac{y}{2y \ln y + y - x}$ **is:**

(A) $x = y^2 (\ln y + c)$ **(B)** $x = y^2 \ln y + cy$
(C) $xy = y^3 \ln y + c$ **(D)** $xy = y^2 \ln y + c$

Sol. **(D)**

$\frac{dy}{dx} = \frac{y}{2y \ln y + y - x}$

or $\frac{dx}{dy} = \frac{2y\ln y + y - x}{y}$ (linear in $\frac{dx}{dy}$)

or $\frac{dx}{dy} + \frac{1}{y}x = (2\ln y + 1)$

I.F. $= y$
Solution is $xy = \int y(2\ln y + 1)dy + c$
$xy = y^2 \ln y + c$

Q. **Suppose that g'(x) exists for $x \in R$ and given that g'(0) = 2 and $g(x+y) = e^y \cdot g(x) + e^x \cdot g(y)$ for all x & y.**
Then which of the following is true
(A) The area bounded by g(x) and the x axis is 3 unit²
(B) g(x) has exactly one local maxima
(C) Greatest value of g(x) is $\frac{2}{e}$
(D) Minimum value of g(x) is $-\frac{2}{e}$

Sol. **(D)**

Put $x = 0$; $y = 0$ $\Rightarrow$ $g(0) = 0$

and $g'(0) = \lim_{h\to 0} \frac{g(h)}{h} = 2$

Now $g'(x) = \lim_{h\to 0} \frac{g(x+h) - g(x)}{h}$

$$= \lim_{h\to 0} \frac{e^h \cdot g(x) + e^x \cdot g(h) - g(x)}{h}$$

$$= g(x) \lim_{h\to 0}\left(\frac{e^h - 1}{h}\right) + e^x \lim_{h\to 0} \frac{g(h)}{h}$$

or $g'(x) = g(x) + 2e^x$

Let $g(x) = y$ then $\frac{dy}{dx} = y + 2e^x$

$\Rightarrow$ $e^{-x} \frac{dy}{dx} + (-e^{-x})\, y = 2$

$\Rightarrow$ $\frac{d}{dx}(y e^{-x}) = 2$

Hence $y = 2x e^x = g(x)$

Now $\frac{dy}{dx} = 2[e^x + x e^x] = 0$ $\Rightarrow$ $x = -1$

$\Rightarrow$ minima at $x = -1$

$\Rightarrow$ range is $\left[-\frac{2}{e}, \infty\right)$

Area $= \int_{-\infty}^{0} 2xe^x = 2$

Q. **The function y (x) satisfies the equation $y(x) + 2x\int_0^x \frac{y(u)}{1+u^2}du = 3x^2 + 2x + 1$.**

Show that the substitution $z(x) = \int_0^x \frac{y(u)}{1+u^2}du$ converts the equation into a first order linear differential equation for z(x), solve for z(x). Hence solve the original equation for y(x).

Sol. $y(x) + 2x\int_0^x \frac{y(u)}{1+u^2}du = 3x^2 + 2x + 1$ ....(1)

Let $z(x) = \int_0^x \frac{y(u)}{1+u^2}du$

$\therefore \quad z'(x) = \frac{y(x)}{1+x^2} \Rightarrow \quad y(x) = (1 + x^2)\, z'(x)$ ....(2)

substituting y (x) in equation (1)

$(1 + x^2)\, z'(x) + 2x \cdot z(x) = 3x^2 + 2x + 1$

Let $z'(x) = \frac{dz}{dx}$

$\frac{dz}{dx} + \frac{2x}{1+x^2}z(x) = \frac{3x^2+2x+1}{1+x^2}$ which is linear DE

I.F. $= e^{\int \frac{2x}{1+x^2}dx} = e^{\ln(1+x^2)} = 1 + x^2$

Hence $z(1 + x^2) = \int (3x^2 + 2x + 1)\,dx$

$z(1 + x^2) = x^3 + x^2 + x + C$

but $z = 0$ when $x = 0 \Rightarrow C = 0$

Hence $z(x) = \frac{x^3 + x^2 + x}{1+x^2}$ .....(3)

now $y(x) = (1 + x^2) \cdot z'(x)$ .....(4) (from 2)

also $z(x) = \frac{x(1+x^2)+x^2}{1+x^2} = x + \frac{x^2+1-1}{1+x^2} = x + 1 - \frac{1}{1+x^2}$

$\therefore \quad z'(x) = 1 + \frac{2x}{(1+x^2)^2}$

$= \frac{(1+x^2)+2x}{(1+x^2)^2}$

$\therefore \quad y(x) = \frac{(1+x^2)^2 + 2x}{(1+x^2)}$ [from eqn (2)]

Mixture Problems :

Differential equation can be used to find a equation which relates amount of salt (or chemcial) present in liquid solution at a given time where solution is kept uniform and additional liquid (may or may not containing salt (chemical) is being run into present mixture).

Differential equation for the above process is based on the concept

$$\text{Rate of change of amount in container} = \begin{pmatrix}\text{rate at which}\\ \text{chemical}\\ \text{arrives}\end{pmatrix} - \begin{pmatrix}\text{rate at which}\\ \text{chemical}\\ \text{departs}\end{pmatrix} \quad ...(1)$$

If we represent amount of salt in container at time t as y (t) and volume at time 't' as V(t), then the departure rate of the chemical at time t is

$$\text{Departure rate} = \frac{y(t)}{V(t)} \cdot (\text{out flow rate})$$

$$= \begin{pmatrix}\text{concentration in}\\ \text{container at time t}\end{pmatrix} \cdot (\text{out flow rate})$$

Accordingly, Equation (1) becomes

$$\frac{dy}{dt} = (\text{chemical's arrival rate}) - \frac{y(t)}{V(t)} \cdot (\text{out flow rate}) \quad ...(2)$$

Examples :

Q. **100 liters of solution containing 50 gms of salt is present in tank. A solution containing 2 gm/liter of salt runs into the tank at the rate of 5 liter/min. The mixture is kept stirring and flows out of the tank at the rate of 4 liters/min then solve the DE of the process and use it to find an expression for the amount of salt present at time t.**

Sol. (a) Rate at which salt is added $= \frac{2\,gm}{liter} \cdot \frac{5\,liters}{min} = 10$ gm/min.

(b) Volume V (t) of solution at time t =initial value + (inflow−outflow).t

$= 100 + (5 - 4)\,t = (100 + t)$ liters

(c) Rate at which salt leaves $= \frac{y(t)}{V(t)}$ (out flow rate) $= \frac{y}{100+t} \cdot 4$

(d) Hence DE is $\frac{dy}{dt} = 10 - \frac{4y}{100+t}$ which is linear $\frac{dy}{dt} + \frac{4y}{100+t} = 10$

$$\text{I.F.} = e^{\int \frac{4}{100+t} dt} = e^{+4(n\,(100+t))} = (100 + t)^4$$

$$\therefore \quad y\,(100 + t)^4 = 10\int (100 + t)^4\,dt + c$$

$y(100+t)^4 = 2(100+t)^5 + c$

if $t = 0 ; y = 50$

$50 \cdot 100^4 = 200 \cdot 100^4 + c$

$\therefore \quad c = -150 \cdot 100^4$

$\therefore \quad y(100+t)^4 = 2(100+t)^5 - 150 \cdot 100^4$

$$y = 2(100+t) - \frac{150 \cdot 100^4}{(100+t)^4} \quad ; \qquad y = 2(100+t) - \frac{150}{\left(1+\frac{t}{100}\right)^4}$$

Q. **A tank contains 200 liters of Solution in which 20 gms of salt dissolved. Solution containing $\frac{1}{4}$ gm of salt/lit. runs into the tank at the rate of 2 liter/ min. The mixture is kept stirring runs out at the same rate.**
Find the limiting value of salt when $t \to \infty$ and also find the time when amount of salt in solution is 20 gm.

Sol. **(i)** $t = 0 \ ; y = 20$

$$\frac{dy}{dt} = \frac{1}{2} - \frac{y}{200} \cdot 2 \quad \Rightarrow \frac{dy}{dt} = \frac{1}{2} - \frac{y}{100} \quad \Rightarrow \quad \frac{50-y}{100} = \frac{dy}{dt}$$

Hence $\frac{dy}{50-y} = \frac{dt}{100}$

$$-\ln(50-y) = \frac{t}{100} - c$$

or $\quad \ln(50-y) = -\frac{t}{100} + c$

$t = 0 ; y = 20 ; c = \ln 30$

Hence $\ln\left(\frac{50-y}{30}\right) = \frac{-t}{100}$

$$\frac{50-y}{30} = e^{-\frac{t}{100}}$$

$$y = 50 - 30 \cdot e^{-\frac{t}{100}}$$

(ii) y when $t \to \infty = 50$ gms

(iii) Never, as $y > 20$ for all $t > 0$

Equations Reducible To Linear Form : (Bernoulli's Equation)

The equation $\frac{dy}{dx} + Py = Q.\ y^n$ where P & Q functions of x, is reducible to the linear form by dividing it by y^n & then substituting $y^{-n+1} = Z$. Its solution can be obtained as in the normal case.

Q. **Solve** $\frac{dy}{dx} + \frac{y}{x} = y^3$

Sol. Dividing the given equation by y^3, we get

$$\frac{1}{y^3}\frac{dy}{dx} + \frac{1}{y^2}\frac{1}{x} = 1 \qquad ...(1)$$

Putting $\frac{1}{y^2} = v$, we have

$$\frac{-2}{y^3}\frac{dy}{dx} = \frac{dv}{dx}$$

$\therefore$ equation (1) becomes

$$-\frac{1}{2}\frac{dv}{dx} + \frac{1}{x}v = 1$$

$$\frac{dv}{dx} - \frac{2}{x}v = -2$$

This is a linear equation with v as the dependent variable.

$$\text{I.F.} = e^{-\int(2/x)dx} = e^{-2\ln x} = e^{\ln\frac{1}{x^2}} = \frac{1}{x^2}$$

Therefore, the solution is

$$\frac{v}{x^2} = \int\frac{-2}{x^2}dx + c$$

$$\frac{v}{x^2} = \frac{2}{x} + c$$

Or $2xy^2 + cx^2y^2 = 1$

Q. **Solution** $\frac{dy}{dx} + xy = xy^2$ **is**

(A) $y = e^{x^2/2} + c$ **(B)** $y = -e^{x^2/2} + c$ **(C)** $y = 2 + ce^{x^2/2}$ **(D)** $y = 1 + ce^{\frac{x^2}{2}}$

Sol. **(D)**

$$\frac{1}{y^2}\frac{dy}{dx} + \frac{1}{y}x = x$$

Let. $\frac{1}{y} = v \Rightarrow -\frac{1}{y^2}\frac{dy}{dx} = \frac{dv}{dx}$

$\Rightarrow \frac{-dv}{dx} + vx = x$

$\Rightarrow \frac{dv}{dx} + (-x)v = -x$

I.F. $= e^{\int -x\,dx} = e^{\frac{-x^2}{2}}$

Solution : $y.e^{\frac{-x^2}{2}} = \int (-x)\, e^{\frac{-x^2}{2}}\, dx + c$

Let $\frac{-x^2}{2} = t \Rightarrow (-xdx) = dt$

$\Rightarrow y.e^{\frac{-x^2}{2}} = \int e^t dt + c$

$\Rightarrow y.e^{\frac{-x^2}{2}} = e^{\frac{-x^2}{2}} + c$

$\Rightarrow y = 1 + c\, e^{\frac{x^2}{2}}$

Q. **Solve the following Differential Equation** $\frac{dy}{dx} = xy + x^3y^2$

Sol. $\frac{dy}{dx} = xy + x^3y^2 \quad \Rightarrow \frac{dy}{dx} - yx = x^3y^2$

$\frac{1}{y^2}\frac{dy}{dx} - \frac{x}{y} = x^3$

Put $t = \frac{-1}{y}$ so $dt = \frac{dy}{y^2}$

$\frac{dt}{dx} + tx = x^3 \Rightarrow$ Integrating factor $= e^{\int x dx} = e^{\frac{x^2}{2}}$

$t\, e^{\frac{x^2}{2}} = \int e^{\frac{x^2}{2}} x^3\, dx = \int e^{\frac{x^2}{2}}\, x\, x^2\, dx - c$

Put $z = \frac{x^2}{2}$; $dz = x\, dx$

$\int e^{\frac{x^2}{2}}\, x^3\, dx = \int e^z (2z) dz = 2(z-1)e^z = (x^2-2)\, e^{\frac{x^2}{2}}$

$te^{\frac{x^2}{2}} = (x^2-2)\, e^{\frac{x^2}{2}} - c$

$$\Rightarrow \frac{-e^{\frac{x^2}{2}}}{y}=(x^2-2)\ e^{\frac{x^2}{2}}-c$$

$$\Rightarrow y=\frac{e^{\frac{x^2}{2}}}{c+(2-x^2)e^{\frac{x^2}{2}}}$$

Q. **Solve $y\sin x\frac{dy}{dx}=\cos x(\sin x-y^2)$**

Sol. Let $z=y^2 \quad \therefore dz=2y\,dy$

$$\frac{1}{2}\frac{dz}{dx}+\cot x\, z=\cos x$$

$$\Rightarrow \frac{dz}{dx}+2\cot x\, z=2\cos x$$

Integrating factor $=e^{\int 2\cot x dx}=e^{2\ln(\sin x)}=\sin^2 x$

$$\therefore y^2\sin^2 x=\int 2\sin^2 x\cos x dx+C$$

$$\therefore y^2\sin^2 x=\frac{2\sin^3 x}{3}+C.$$

Q. **Solution of $x\frac{dy}{dx}+y\ln y=xye^x$ is:**

(A) $y\ln y=x-1+ce^{-x}$ **(B) $x\ln y=(x+1)e^x+c$**

(C) $y\ln x=(x+1)e^x+c$ **(D) $x\ln y=e^x(x-1)+c$**

Sol. **(D)**

Divide by xy

$$\frac{1}{y}\frac{dy}{dx}+\ln y.\frac{1}{x}=e^x$$

Let $\ln y=v \Rightarrow \frac{1}{y}dy=dv$

$$\frac{dv}{dx}+\frac{v}{x}=e^x \qquad \text{(linear)}$$

$$\text{I.F.}=e^{\int\frac{dx}{x}}=e^{\ln(x)}=x$$

Solution is $x.v=\int x.e^x dx+c$

$x\ln y=xe^x-\int e^x dx+c$ (Applying by parts)

$x\ln y=xe^x-e^x+c$

Q. **Solution of $\sec^2 y\frac{dy}{dx}+2x\tan y = x^3$ is:**

(A) $\tan y = x^2+1-ce^{-x^2}$ **(B)** $\tan y = x^2-1+ce^{-x^2}$

(C) $2e^{x^2}\tan y = e^{x^2}(x^2-1)+2c$ **(D)** $2\tan y = x^2+1+ce^{-x^2}$

Sol. **(C)**

Let $\tan y = v \Rightarrow \sec^2 y\frac{dy}{dx} = \frac{dv}{dx}$

$\frac{dv}{dx}+2xv = x^3$ (linear)

I.F. $= e^{\int 2xdx} = e^{x^2}$

Solution is $ve^{x^2} = \int e^{x^2}.x^3dx+c$

Let $x^2 = t \Rightarrow 2xdx = dt$

$e^{x^2}\tan y = \frac{1}{2}\int e^t tdt+c$

$e^{x^2}\tan y = \frac{e^t.t-\int e^t}{2}+c$

$\Rightarrow (\tan y)e^{x^2} = \frac{1}{2}e^t(t-1)+c$

$\Rightarrow 2e^{x^2}\tan y = e^{x^2}(x^2-1)+2c$

EXACT DIFFERENTIALS

Note: Following Exact Differentials Must Be Remembered :

(i) $xdy+ydx = d(xy)$

(ii) $\frac{xdy-ydx}{x^2} = d\left(\frac{y}{x}\right)$

(iii) $\frac{ydx-xdy}{y^2} = d\left(\frac{x}{y}\right)$

(iv) $\frac{xdy+ydx}{xy} = d(\ln xy)$

(v) $\frac{dx+dy}{x+y} = d(\ln(x+y))$

(vi) $\frac{xdy-ydx}{xy} = d\left(\ln\frac{y}{x}\right)$

(vii) $\frac{ydx-xdy}{xy} = d\left(\ln\frac{x}{y}\right)$

(viii) $\frac{xdy-ydx}{x^2+y^2} = d\left(\tan^{-1}\frac{y}{x}\right)$

(ix) $\frac{ydx-xdy}{x^2+y^2} = d\left(\tan^{-1}\frac{x}{y}\right)$

(x) $\dfrac{x\,dx + y\,dy}{x^2 + y^2} = d\left[\ln\sqrt{x^2 + y^2}\right]$ **(xi)** $d\left(-\dfrac{1}{xy}\right) = \dfrac{x\,dy + y\,dx}{x^2 y^2}$

(xii) $d\left(\dfrac{e^x}{y}\right) = \dfrac{y\,e^x\,dx - e^x\,dy}{y^2}$ **(xiii)** $d\left(\dfrac{e^y}{x}\right) = \dfrac{x\,e^y\,dy - e^y\,dx}{x^2}$

Q. **Solve $xdx + ydy = \dfrac{xdy - ydx}{x^2 + y^2}$**

Sol. The D.E. can be written as

$\dfrac{1}{2}d(x^2+y^2) = d\{\tan^{-1}\left(\dfrac{y}{x}\right)\}$

Integrating we get

$\dfrac{1}{2}(x^2+y^2) = \tan^{-1}\left(\dfrac{y}{x}\right) + c$

Q. **Solve $\{(x+1)\left(\dfrac{y}{x}\right) + \text{siny}\}\,dx + (x + \log x + x\cos y)\,dy = 0$**

Sol. We can re-write the differential equation

$(y\,dx + x\,dy) + \left(\dfrac{y}{x}dx + \log x\,dy\right) + (\text{siny}\,dx + x\cos y\,dy) = 0$

$\Rightarrow d(xy) + d(y\log x) + d(x\sin y) = 0$

Integrating both the sides we have

$xy + y\log x + x\sin y = c$

Q. **The general solution of the differential equation $x\,dy + y\,dx + 2x^3\,dx = 0$, is**

(A) $xy + x^4 = C$ **(B)** $xy + \dfrac{1}{2}x^4 = C$ **(C)** $x + y + \dfrac{1}{2}x^4 = 0$ **(D)** $xy - \dfrac{1}{2}x^4 = C$

Sol. **(B)**

$(xdy + ydx) + 2x^3dx = 0$

$\Rightarrow d(xy) = -2x^3dx$

$\Rightarrow \int d(xy) = \int -2x^3dx$

$\Rightarrow xy = \dfrac{-2x^4}{4} + C$

$\Rightarrow xy = \dfrac{-x^4}{2} + C$

Q. **Solve this differential equation $(y+ \sin x \cos^2(xy))dx + xdy + \sin y \cos^2(xy)\, dy = 0$**

Sol.

$$\frac{x\,dy + y\,dx}{\cos^2 xy} + \sin x\,dx + \sin y\,dy = 0$$

$$\int \frac{d(xy)}{\cos^2 xy} + \int \sin x\,dx + \int \sin y\,dy = c$$

$$\tan(x\,y) = \cos x + \cos y + c$$

Q. **Solution of the differential education $\left\{\frac{1}{x} - \frac{y^2}{(x-y)^2}\right\}dx + \left\{\frac{x^2}{(x-y)^2} - \frac{1}{y}\right\}dy = 0$ is**

(A) $\ln\left|\frac{x}{y}\right| + \frac{xy}{x-y} = c$ **(B)** $\frac{xy}{x-y} = ce^{x/y}$

(C) $\ln|xy| = c + \frac{xy}{x-y}$ **(D) None of these**

Sol. **(A)**

$$\left\{\frac{1}{x} - \frac{y^2}{(x-y)^2}\right\}dx + \left\{\frac{x^2}{(x-y)^2} - \frac{1}{y}\right\}dy = 0$$

$$\Rightarrow \left(\frac{dx}{x} - \frac{dy}{y}\right) + \left(\frac{x^2dy - y^2dx}{(x-y)^2}\right) = 0$$

$$\Rightarrow \left(\frac{dx}{x} - \frac{dy}{y}\right) + \left(\frac{dy/y^2 - dx/x^2}{(1/y - 1/x)^2}\right) = 0$$

Integrating we get in $\ln|x| - \ln|y| + \frac{1}{(1/y - 1/x)} = c$

$$\Rightarrow \ln\left|\frac{x}{y}\right| + \frac{xy}{x-y} = c$$

Q. **The solution of the differential equation $\left\{1 + x\sqrt{(x^2+y^2)}\right\}dx + \left\{\sqrt{(x^2+y^2)} - 1\right\}y\,dy = 0$ is**

(A) $x^2 + \frac{y^2}{2} + \frac{1}{3}(x^2+y^2)^{3/2} = c$ **(B)** $x^2 - \frac{y^3}{2} + \frac{1}{2}(x^2+y^2)^{1/2} = c$

(C) $x - \frac{y^2}{2} + \frac{1}{6}(x^2+y^2)^{3/2} = c$ **(D)** $x - \frac{y^2}{2} + \frac{1}{3}(x^2+y^2)^{3/2} = c$

Sol. **(D)**

$$\left\{1 + x\sqrt{(x^2 + y^2)}\right\} dx + \left\{\sqrt{(x^2 + y^2)} - 1\right\} y\, dy = 0$$

$$\Rightarrow dx - y\, dy + \sqrt{(x^2 + y^2)}\ (x dx + y\, dy) = 0$$

$$\Rightarrow dx - y\, dy + \frac{1}{2}\sqrt{(x^2 + y^2)}\, d(x^2+y^2) = 0$$

Integrating we have

$$x - \frac{y^2}{2} + \frac{1}{2}\int \sqrt{t}\, dt = c, \quad \left\{t = \sqrt{(x^2 + y^2)}\right\}$$

$$\text{Or } x - \frac{y^2}{2} + \frac{1}{3}(x^2 + y^2)^{3/2} = c$$

Trajectories :

Suppose we are given the family of plane curves.

ϕ (x, y, a) = 0

depending on a single parameter a.

A curve making at each of its points a fixed angle α with the curve of the family passing through that point is called an isogonal trajectory of that family ; if in particular $\alpha = \pi/2$, then it is called an orthogonal trajectory.

Orthogonal trajectories :

For differential equation of form F(x,y,y') = 0,

the equation F(x, y, -1/y') = 0 is a orthogonal trajectory

In order to find orthogonal trajectory for a given curve, find its differential equation and replace y' by -1/y' and solve the new equation to get orthogonal trajectory for the 1st curve

EXAMPLES:

Q. **The orthogonal trajectories of the family xy = c is:**
(A) $x^2-y^2 = k$ **(B) $x^2 + y^2 = k$** **(C) $x^3-y^3 = k$** **D) $x^4 + y^4 = k$**

Sol. **(A)**

$xy = c$

Differentiating w.r.t. x, we get $x\frac{dy}{dx} + y = 0$

Replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$ we get $x\frac{dx}{dy} - y = 0$

$x\,dx - y\,dy = 0$

Integrating

$\Rightarrow x^2 - y^2 = k$

This is the family of required orthogonal trajectories.

Q. **The equation of all possible curves that will cut each member of the family of the curves $x^2 + y^2 - 2cy = 0$ at right angle, is:**

(A) $y = a(x + a)$ **(B) $x^2 + y^2 = ax$** **(C) $y = ax$** **(D) $x^2 + y^2 = a^2$**

Sol. **(B)**

$x^2 + y^2 - 2cy = 0$

$\Rightarrow 2x + 2yy' - 2cy' = 0$

$\Rightarrow 2c = \left(\frac{2x + 2yy'}{y'}\right)$

$\therefore x^2 + y^2 - \left(\frac{2x + 2yy'}{y'}\right)y = 0$

For orthogonal trajectory, replace y' by $-\frac{1}{y'}$

$$x^2 + y^2 - \frac{\left(2x - \frac{2y}{y'}\right)y}{\left(\frac{-1}{y'}\right)} = 0 \Rightarrow x^2 + y^2 + (2xy' - 2y)y = 0 \Rightarrow x^2 + y^2 + 2xy\frac{dy}{dx} - 2y^2 = 0$$

$\frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$ (a homogeneous differential equation)

Put $y = vx \quad \therefore \frac{dy}{dx} = v + x\frac{dv}{dx}$

$\Rightarrow v + x\frac{dv}{dx} = \frac{v^2 - 1}{2v} \quad \Rightarrow x\frac{dv}{dx} = \frac{v^2 - 1}{2v} - v = \frac{-(v^2 + 1)}{2v} \quad \Rightarrow \frac{2v}{v^2 + 1}dv = \frac{-dx}{x}$

$\Rightarrow \ln(v^2 + 1) = -\ln x + \ln a \Rightarrow x^2 + y^2 = ax$

Q. **Which of the following is the orthogonal trajectory of the family $y^2 = 4k(x + k)$ is:**

(A) $x + \sqrt{x^2 + y^2} = cy^2$ **(B)** $x + \sqrt{x^2 + y^2} = c$

(C) Both A and B **(D) None of these**

Sol. **(C)**

$2 y y_1 = 4k$ ...(1)

$$y^2 = 2 y y_1 \left(x + \frac{y y_1}{2}\right)$$

$$y^2 = 2x y y_1 + y^2 \cdot y_1^2$$

$$y^2 \left(\frac{dy}{dx}\right)^2 + 2xy \left(\frac{dy}{dx}\right) - y^2 = 0$$

$$y \left(\frac{dy}{dx}\right)^2 + 2x \left(\frac{dy}{dx}\right) - y = 0$$

For orthogonal trajectory, replace y' by $-\frac{1}{y'}$

$$y \left(\frac{dx}{dy}\right)^2 - 2x \left(\frac{dx}{dy}\right) - y = 0$$

$$\frac{dx}{dy} = \frac{2x \pm \sqrt{4x^2 + 4y^2}}{2y} = \frac{x \pm \sqrt{x^2 + y^2}}{y}$$

Let $x = vy \Rightarrow \frac{dx}{dy} = v + y \frac{dv}{dy}$

$$v + y \frac{dv}{dy} = v \pm \sqrt{1 + v^2}$$

$$\int \frac{dv}{\sqrt{1 + v^2}} = \pm \int \frac{dy}{y}$$

$\ln (v + \sqrt{1 + v^2}) = \ln(c) \pm \ln (y)$

Case I (+ sign)

$\ln (v + \sqrt{1 + v^2}) = \ln (c y)$

$$\frac{x + \sqrt{x^2 + y^2}}{y} = cy$$

$x + \sqrt{x^2 + y^2} = cy^2$ Ans.

Case II (– sign)

$$\ln\left(v+\sqrt{1+v^2}\right)=\ln(c)-\ln(y)$$

$$\frac{x+\sqrt{x^2+y^2}}{y}=\frac{c}{y}$$

$$x+\sqrt{x^2+y^2}=c$$

MISCELLANEOUS

1. SPECIAL Type y f (xy) dx + xg (xy) dy = 0

In an equation of the form : y f (xy) dx + x g (xy) dy = 0 the variables can be separated by the substitution xy = v.

Q. $\dfrac{ydx}{x^3y^3-x^2y^2-xy+1}+\dfrac{xdy}{x^3y^3+x^2y^2+xy+1}=0$

Sol. $xy=v \Rightarrow y=\dfrac{v}{x}$

$$x\frac{dy}{dx}+y=\frac{dv}{dx} \qquad ...(1)$$

now equation become $y.(v^3+v^2+v+1)+(v^3-v^2-v+1)\,x.\dfrac{dy}{dx}=0$

$$y.(v^3+v^2+v+1)+(v^3-v^2-v+1)\left[\frac{dv}{dx}-y\right]=0$$

$$y.\left[(v^3+v^2+v+1)-(v^3-v^2-v+1)\right]+\frac{dv}{dx}(v^3-v^2-v+1)=0$$

$$\frac{v}{x}(2v^2+2v)+\frac{dv}{dx}(v^3-v^2-v+1)=0$$

$$\frac{2v^2(v+1)}{x}+\frac{dv}{dx}(v^3-v^2-v+1)=0$$

$$\Rightarrow 2\int\frac{dx}{x}+\int\frac{v^3-v^2-v+1}{v^2(v+1)}dv=0$$

$$2\int\frac{dx}{x}+\int\frac{(v^2-1)(v-1)}{v^2(v+1)}dv=0$$

$$2\int\frac{dx}{x}+\int\frac{v^2-2v+1}{v^2}dv=0$$

$$\Rightarrow 2\ln(x)+v-2\ln v-\frac{1}{v}=c,$$

where v = xy.

Q. **Solution of $y(xy + e^{xy})dx + x(xy - e^{xy})dy = 0$ is**

(A) $\ln y - \ln x = e^{-xy} + c$ **(B)** $\ln xy + e^{-xy} = c$

(C) $2\ln x - \ln y = e^{-xy} + c$ **(D)** $\ln x - \ln y = c + e^{-xy}$

Sol. **(D)**

Let $xy = v \Rightarrow \frac{dy}{dx} = \frac{xdv - vdx}{x^2}$

Given equation become

$$v(v+e^v) + (v-e^v)\left(x\frac{dv}{dx} - v\right) = 0$$

$$2ve^v = (e^v - v)x\frac{dv}{dx}$$

$$\frac{e^v - v}{ve^v}dv = 2\frac{dx}{x}$$

$$2\frac{dx}{x} = \frac{dv}{v} - \frac{dv}{e^v}$$

$$\Rightarrow \int\frac{2dx}{x} = \int\frac{dv}{v} - \int e^{-v}dv$$

$\Rightarrow 2\ln x = \ln v + e^{-v} + c$

$\Rightarrow \ln(x) - \ln(y) = e^{-xy} + c$ Putting value of v

2. Misleading appearance as Homogeneous

A D. E. containing terms like $\frac{y}{x}$ and $\frac{x}{y}$ may not be homogeneous and this can not be solved by substituting $y = v\,x$.

Q. $$\left(\frac{1}{y}\sin\frac{x}{y} - \frac{y}{x^2}\cos\frac{y}{x} + 1\right)dx + \left(\frac{1}{x}\cos\frac{y}{x} - \frac{x}{y^2}\sin\frac{x}{y} + \frac{1}{y^2}\right)dy = 0$$

Sol. Isolating $\sin\frac{x}{y}$ & $\cos\frac{y}{x}$

$$\left(\frac{dx}{y} - \frac{x\,dy}{y^2}\right)\sin\frac{x}{y} + \left(\frac{dy}{x} - \frac{y\,dx}{x^2}\right)\cos\frac{y}{x} + dx + \frac{dy}{y^2} = 0$$

$$\frac{ydx - xdy}{y^2}.\sin\frac{x}{y} + \frac{xdy - ydx}{x^2}.\cos\frac{y}{x} + dx + \frac{dy}{y^2} = 0$$

$$\sin\frac{x}{y}.d\left(\frac{x}{y}\right) + \cos\frac{y}{x}.d\left(\frac{y}{x}\right) + dx + \frac{dy}{y^2} = 0$$

$$-\cos\frac{x}{y}+\sin\frac{y}{x}+x-\frac{1}{y}=c$$

Q. $\left(x\cos\frac{y}{x}+y\sin\frac{y}{x}\right)ydx+\left(x\cos\frac{y}{x}-y\sin\frac{y}{x}\right)xdy=0$

Sol. This differential equation is Homogeneous and can be solved by substitution $y = vx$.

put $y = vx \Rightarrow \frac{dy}{dx}=v+x\frac{dv}{dx}$

Given equation becomes $(\cos v+v\sin v)v+(\cos v-v\sin v)\left(v+x\frac{dv}{dx}\right)=0$

simplifies to $\int\frac{\cos v-v\sin v}{v\cos v}dv=-\int\frac{2dx}{x}$

$$\int\left(\frac{1}{v}-\frac{\sin v}{\cos v}\right)dv=-\int\frac{2dx}{x}$$

$\therefore$ ln (v) + ln (cosv) =–2 ln x + ln c

$\therefore$ ln (v cosv) + ln x^2 = ln c

$\Rightarrow xy\cos\frac{y}{x}=c$

Q. $\frac{dy}{xsiny+ycosy}+\frac{dx}{xcosy-ysiny}=0$

Sol. Let x siny + y cosy = t

differentiate w.r.t. x

$$x\cos y\frac{dy}{dx}+\sin y+\cos y\frac{dy}{dx}-y\sin y\frac{dy}{dx}=\frac{dt}{dx}$$

$$(x\cos y-y\sin y)\frac{dy}{dx}+\sin y+\cos y\frac{dy}{dx}=\frac{dt}{dx} \qquad ...(1)$$

Put in given equation

$$\frac{dt}{dx}-\left(\sin y+\cos y\frac{dy}{dx}\right)+t=0$$

$$\sin y=z\ ;\ \cos y\frac{dy}{dx}=\frac{dz}{dx}$$

$$\frac{dt}{dx}-\left(z+\frac{dz}{dx}\right)+t=0 \quad \text{or} \quad \frac{d}{dx}(t-z)+(t-z)=0$$

Let $t - z = u$

$\frac{du}{dx} + u = 0$

$\Rightarrow \ln u + x = C$

$\Rightarrow \ln (t - z) + x = C$

$\ln (x \sin y + y \cos y - \sin y) = C - x$

$\Rightarrow x \sin y + y \cos y - \sin y = c\, e^{-x}$

Q. **Prove the identity** $\int_0^x e^{zx-z^2} dz = e^{x^2/4} \int_0^x e^{-z^2/4} dz$

deriving for the function I (x) = $\int_0^x e^{zx-z^2} dz$ **a D.E. and solving it.**

Sol: $I(x) = \int_0^x e^{zx-z^2} dz \quad \left[zx - z^2 = \frac{x^2}{4} - \left(z - \frac{x}{2}\right)^2 \right]$

$\therefore \quad I(x) = e^{x^2/4} \int_0^x e^{-\left(z-\frac{x}{2}\right)^2} dz$

put $z - \frac{x}{2} = t$, $dz = dt$

$= e^{x^2/4} \int_{-\frac{x}{2}}^{\frac{x}{2}} e^{-t^2} dt$

$I(x) = 2.e^{x^2/4} \int_0^{\frac{x}{2}} e^{-t^2} dt \qquad ...(1)$ (f (t) is an even function)

Diff. (1) with respect to x.

$$\frac{dI}{dx} = 2\left[e^{\frac{x^2}{4}} . e^{-\frac{x^2}{4}} . \frac{1}{2} + \left(\int_0^{\frac{x}{2}} e^{-t^2} dt \right) . e^{\frac{x^2}{4}} . \frac{x}{2} \right] = 1 + x\, e^{\frac{x^2}{4}} \int_0^{\frac{x}{2}} e^{-t^2} dt = 1 + x . \frac{I}{2}$$

$\frac{dI}{dx} - \frac{x}{2} I = 1$

$IF = e^{-\frac{x^2}{4}}$

$\therefore \quad I \cdot e^{-\frac{x^2}{4}} = \int e^{-\frac{x^2}{4}} dx + C = F(x) + C$ (say, where $\int e^{-\frac{x^2}{4}} dx = F(x)$)

when $x = 0$ then $I = 0$

$\Rightarrow \quad C = -F(0)$

$$I \cdot e^{-\frac{x^2}{4}} = F(x) - F(0) = \int_0^x e^{-\frac{z^2}{4}} dz$$

$$\therefore \quad I = e^{\frac{x^2}{4}} \cdot \int_0^x e^{-\frac{z^2}{4}} dz = \text{R.H.S.}$$